

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

PROBLEM-SOLUTION FIT CANVAS

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Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Computer science students, technical trainees, or web development learners who need to master, practice, and articulate complex concepts verbally for technical interviews or assessments.	6. CUSTOMER LIMITATIONS [CL] Limited access to personalized, expert mentors for speaking feedback; high latency or cost in cloud API processing; constraint of relying initially on local desktop environments without high-end dedicated GPUs.
2. PROBLEMS / PAINS & FREQUENCY [PR] Difficulty identifying conceptual gaps or technical errors in explanations, and frequent anxiety or lack of fluency (tracked during every explanation practice block).	5. AVAILABLE SOLUTIONS, PROS & CONS [AS] Mock Interviews with Peers: Pros: Interactive. Cons: Unreliable metrics, highly inconsistent availability. Traditional LMS Quizzes: Pros: Instant grading. Cons: Text-only focus; completely fails to measure verbal articulation or speech fluency.
3. TRIGGERS TO ACT [TR] Approaching technical job interviews, comprehensive web development assessments, or the need to self-verify core competency scores in real-time.	7. BEHAVIOR & INTENSITY [BE] High intensity of frustration when learners repeat core concepts but cannot measure progress or fluency objectively, leading to vocal filler over-use (um, uh) and irregular speech pacing.
4. EMOTIONS BEFORE / AFTER [EM] Before: Anxious, uncertain, and unconfident regarding technical explanation skills. After: Validated, clear-headed, and motivated by granular score performance sheets and comprehensive generative feedback.	8. CHANNELS OF BEHAVIOR [CH] Online: Independent practice sessions through Streamlit UI apps, GitHub testing environments. Offline: Verbal study groups, classroom technical explanations, mock presentations.
9. PROBLEM ROOT / CAUSE [RC]	10. YOUR SOLUTION [SL]

Conventional education systems rely exclusively on rote written verification (MCQs/code snippets), completely overlooking verbal conceptual explanation testing as a core professional metric.

VBCUA: An integrated ML-powered speech evaluation pipeline combining Whisper transcription, Sentence-BERT semantic similarity check (60%), Librosa speech fluency tracking (40%), and Google Gemini dynamic criticism generation.